Homework 14: Unitary Matrices and Linear Transformations

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Definition. ***Unitary Matrix.***

If $F = \mathbb{R}$ or $F = \mathbb{C}$ then a matrix $U \in M_n(F)$ is called a unitary matrix if its columns form an orthonormal basis of F^n (with respect to the usual dot product). For example, the identity matrix is unitary since its columns are the standard basis vectors $\boldsymbol{e_1}, \boldsymbol{e_2}, \dots, \boldsymbol{e_n}$, which form the standard basis of F^n, which is orthonormal.

As a slightly less trivial example, consider the matrix

$$U = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

which we can show is unitary simply by noting that

$$\left\{\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\}$$

(i.e., the set consisting of the columns of U) is an orthonormal basis of $\mathbb{R}^2$.

Theorem 1. ***Characterization of Unitary Matrices.***

Suppose $F = \mathbb{R}$ or $F = \mathbb{C}$ and $U \in M_n(F)$. The following are equivalent:

a) U is unitary,

b) U^* is unitary,

c) $UU^* = I$,

d) $U^*U = I$,

e) $(U\boldsymbol{v}) \cdot (U\boldsymbol{w}) = \boldsymbol{v} \cdot \boldsymbol{w}$ for all $\boldsymbol{v}, \boldsymbol{w} \in F^n$, and

f) $\|U\boldsymbol{v}\| = \|\boldsymbol{v}\|$ for all $\boldsymbol{v} \in F^n$.

Problem 1. ***Rotation Matrices are Unitary.***

Recall from lectures right before sessional I examination that the standard matrix of the linear transformation

$$R_\theta: \mathbb{R}^2 \to \mathbb{R}^2$$

that rotates $\mathbb{R}^2$ counter-clockwise by an angle of θ is

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

Show that R_θ is unitary.

Hint. Use one of the equivalent statements from theorem stated above.

Problem 2. ***Reflections.***

Recall from introductory linear algebra that the standard matrix of the linear transformation $F_{\boldsymbol{u}}: \mathbb{R}^n \to \mathbb{R}^n$ that reflects $\mathbb{R}^n$ through the line in the direction of the unit vector $\boldsymbol{u} \in \mathbb{R}^n$ is

$$F_{\boldsymbol{u}} = 2\boldsymbol{u}\boldsymbol{u}^T - I\,.$$

Show that $F_{\boldsymbol{u}}$ is unitary.

Elaboration. ***Standard Matrices of Reflections.***

Find the standard matrices of the linear transformations that reflect through the lines in the direction of the following vectors, and depict these reflections geometrically.

a) $\boldsymbol{u} = (0,1) \in \mathbb{R}^2$, and

b) $\boldsymbol{w} = (1,1,1) \in \mathbb{R}^3$.

Recall from lectures before sessional I that the matrix of reflection through the $y-$axis is

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Solution.

a). Since $\boldsymbol{u}$ is a unit vector, the standard matrix of F_u is simply

$$F_{\boldsymbol{u}} = \boldsymbol{2uu^T} - I = 2\begin{bmatrix} 0 \\ 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 2\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Geometrically, this matrix acts as a reflection through the $y-$axis.

We can verify our computation of $F_{\boldsymbol{u}}$ by noting that multiplication by it indeed just flips the sign of the $x-$entry of the input vector:

$$F_{\boldsymbol{u}}(\boldsymbol{v}) = F_{\boldsymbol{u}}\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix}.$$

b). Since $\boldsymbol{w}$ is not a unit vector, we have to first normalize it (i.e., divide it by its length) to turn it into a unit vector pointing in the same direction.

$$||\boldsymbol{w}|| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}\,,$$

So, we let

$$\boldsymbol{u} = \frac{\boldsymbol{w}}{||\boldsymbol{w}||} = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

The standard matrix of $F_{\boldsymbol{u}}$ is then

$$F_{\boldsymbol{u}} = 2\boldsymbol{u}\boldsymbol{u}^T - I = 2 \cdot \frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}.$$

In general, if $\boldsymbol{u} \in \mathbb{R}^n$ is a unit vector then $F_{\boldsymbol{u}}(\boldsymbol{u}) = \boldsymbol{u}$: $F_{\boldsymbol{u}}$ leaves the entire line in the direction of $\boldsymbol{u}$ alone.

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Good Luck